1. Process of Connecting Tableau to Multiple Data Sources

In Tableau, we often need to use more than one data source to perform a complete analysis. Tableau supports connecting to multiple data sources like Excel, CSV, SQL databases, Google Sheets, and many more.

Steps to connect multiple data sources:

1. **Start Tableau Desktop** and connect to your **primary data source** (for example, Excel or a SQL database).
2. Once connected, the data is added to the **Data pane**.
3. To add another data source, click **"Data" > "New Data Source"** from the top menu.
4. Choose the second data source (for example, a CSV file or another database).
5. Tableau now has both data sources available in the Data pane.
6. You can create **Joins** (if both sources are from the same database) or **Data Blending** (when the sources are different).
   - **Data blending** is done using a **common field** (called a linking field), and Tableau automatically shows a small **orange link icon** when the sources are connected.
   - The first data source used in a worksheet becomes the **primary source** (with a blue checkmark), and the second becomes the **secondary source**.

Why it's important:

Connecting to multiple data sources allows for deeper analysis by combining different types of information. For example, you could blend customer sales data (from Excel) with demographic data (from a database) to analyze how age or location affects purchasing behavior.

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2. Metadata in Tableau

Metadata is "data about data." It helps users understand the structure, type, and meaning of data fields in Tableau.

Components of metadata in Tableau:

- Field names and their types (e.g., string, number, date)
- Whether the field is a **dimension** or a **measure**
- Aggregation behavior (SUM, AVG, etc.)
- Field descriptions and comments
- Calculated fields
- Default formats and sorting rules

Why metadata is useful:

- It provides **clarity** on how each field behaves.
- It ensures **data accuracy** when creating visualizations.
- It helps **document the data model** and calculations, especially in team projects.
- With metadata, Tableau users can **quickly understand** and use fields without looking into the raw data.

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3. Significance of Calculated Fields and Table Calculations in Tableau (with example)

Calculated Fields:

A **calculated field** allows users to create new data fields using formulas and expressions based on the existing data.

Example:

```
tableau
CopyEdit
Profit Ratio = [Profit] / [Sales]
```

This calculated field helps users understand how much profit is generated from sales.

Significance:

- Enables **custom metrics** specific to business needs.
- Reduces the need to modify the original data source.
- Makes dashboards more powerful and dynamic.

Table Calculations:

These are **calculations performed on the results shown in the view**, such as running totals, moving averages, rankings, etc.

Example:

```
tableau
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RUNNING_SUM([Sales])
```

This displays the cumulative sales over time.

Significance:

- Great for **trend analysis** and advanced analytics.
- Used when calculations depend on the **layout of the view**.
- Helps users compare data across dimensions like time, product, or region.

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4. Context Filters and Quick Filters

Context Filters:

A **context filter** is used to set the context for all other filters. When multiple filters are used, the context filter is applied first. This improves performance and controls the order of filter execution.

Example:

If you have filters for "Region" and "Product Category", and you set "Region" as a context filter, then "Product Category" will only show values relevant to that region.

Quick Filters (Normal Filters):

Quick filters are user-friendly filters that appear on the worksheet or dashboard. They allow users to easily include or exclude data based on selected values.

Key Difference:

- **Context filter**: Applied first, affects other filters.
- **Quick filter**: Applied after context filter, can be dependent on it.

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5. Why is the Order of Filtering Important?

In Tableau, filters are executed in a specific order, and this impacts the final result.

Order of Filter Execution:

1. **Extract filters**
2. **Data source filters**
3. **Context filters**
4. **Top N filters**
5. **Dimension filters**
6. **Measure filters**
7. **Table calculations**
8. **Rendering the view**

Why it matters:

- If a filter is applied after aggregation, it may not affect the raw data.
- Some calculations (like table calculations) are done **after filters**, so applying a measure filter **before** the table calculation can produce incorrect results.
- **Context filters** affect what data is available for other filters and can significantly change the analysis.

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6. Design a Tableau Dashboard that Uses Multiple Worksheets and Filters

Example Design:

Goal: Analyze company sales performance across regions and products.

Worksheets:

1. **Map View** – Sales by Region (shows total sales on a geographic map)
2. **Bar Chart** – Sales by Product Category
3. **Line Chart** – Monthly Sales Trend over the last year

Filters Used:

- **Date Filter**: Allows selecting a date range.
- **Region Filter**: Affects all worksheets to filter by selected region.
- **Category Filter**: Filters product categories.

Dashboard Features:

- Use **filter actions** to allow clicking on a region in the map to update the bar chart and line chart.
- Add **dropdown filters** to the sidebar for interactivity.
- Use **parameters** to let the user choose between metrics like "Sales", "Profit", or "Quantity".

Purpose:

This type of dashboard gives a complete overview of sales by region, product performance, and time-based trends – all in one interactive interface.

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7. Types of Calculations in Tableau with Examples

Tableau supports various types of calculations that help in data analysis and visualization:

a. Numeric Calculations

Used to perform arithmetic operations.

Example:
`Profit Margin = [Profit] / [Sales]`

b. String Calculations

Used for text manipulation like combining fields, replacing text, etc.

Example:
`Full Name = [First Name] + " " + [Last Name]`

c. Date Calculations

Used for working with date and time fields.

Example:
`Order Year = YEAR([Order Date])`

Can be used to group or filter data by year, quarter, or month.

d. Table Calculations

These operate on the results shown in the view (after aggregation). They are used for advanced calculations like rankings, moving averages, and percent differences.

Example:
`RANK([Sales])` – Gives the sales rank for each item.

e. Level of Detail (LOD) Expressions

Used when we want to calculate values at a different level of detail than what is shown in the view.

Types of LOD:

- **FIXED** – Calculates at a fixed level, regardless of filters.

  Example:
  `FIXED [Customer ID]: SUM([Sales])`

- **INCLUDE** – Adds additional detail to the calculation.

  Example:
  ```
  INCLUDE [Product]: AVG([Sales])
  ```

- **EXCLUDE** – Removes a level of detail from the calculation.

  Example:
  ```
  EXCLUDE [Product]: SUM([Sales])
  ```

Why they matter:

- LODs help perform precise calculations even when filters or dimensions are involved.
- They solve many complex problems that regular aggregates and filters can't.